

Artificial Intelligence and Technology for Improving the Efficiency of The Judiciary in Nigeria

Eli Adama Jiya^{1,*}; Ahmed Nuruddeen Samaila²; Yusuf Surajo³

¹Department of Computer Science, Federal University Dutsinma, Katsina State, Nigeria.

²Department of Software Engineering, Federal University Dutsinma, Katsina State, Nigeria.

³Department of Cyber Security, Federal University Dutsinma, Katsina State, Nigeria.

Corresponding Author*

DOI: <https://doi.org/10.51584/IJRIAS.2023.8509>

Received: 25 March 2023; Accepted: 05 April 2023; Published: 13 June 2023

Abstract: Technology is an essential component of the civilization of twenty-first-century society and one of the key forces influencing public sector organizations. Public infrastructures like transportation systems and important service delivery are managed using technology and powerful computing software. Artificial intelligence (AI) on the other hand helps to add intelligence to computing devices and applications through learning, reasoning, and perceiving things from data. However, the level of adoption or application of these technologies in the Nigerian judiciary seems to be low as seen during the covid-19 outbreak. The closed down of courts led to an increase in the backlog of cases due to the suspension of court proceedings. Even the option of filing cases using technology like WhatsApp could not function effectively. AI can This paper identifies some areas technology and AI can improve the performance of Nigeria's judiciary and increase the speed of justice dispensation.

Keywords: Artificial Intelligence in Judiciary, Information Technology in Judiciary, Artificial Intelligence and Legal System, Legal System in COVID-19

I. Introduction

Technology is the binding thread for the civilization of the twenty-first century, and it is transforming the planet in all spheres (Contini, 2020). Today, one of the key forces influencing public sector organizations is emerging technologies. These technologies are used to handle public infrastructures and service delivery. The drive from manual systems to online service delivery is a key agenda of many countries around the globe and was the saving option for many developed worlds during the covid-9 lockdown. The use of technology was also essential in aiding some developed nations to even continue court proceeding online (Sourdin et al., 2018)

Artificial intelligence on the other hand has also significantly transformed the world of computing by enabling devices to learn, reason, and perceive things from data and their environment (Xu, 2022). This level of intelligence of machines is made possible by machine learning algorithms that provide intelligent analysis of massive data (Contini, 2020). Artificial intelligence (AI) has already begun to have a significant impact on the field of law, and it is likely to continue to do so in the future (Journal & Sourdin, 2018). Judicial systems are using Artificial Intelligence (AI) to analyse large amounts of legal data to help lawyers identify precedents in case law. The gains of AI also support judges with predictions on issues including sentence duration and recidivism scores.

Apart software systems, AI boots and robots are beginning to transform legal practice in the form lawyers. A good instance is Robot Ross by Watson, IBM's cognitive computer. The robot demonstrated capacity to speak and interact with human language and to act as an intelligent and efficient consultant in bankruptcy cases. According (Lu & Hu, 2019), a robot called DoNotPay developed in UK, can help users in traffic related cases and to prepare necessary papers to defend their case. Another case is a legal robot developed in Israel which went in contest with 20 experience lawyers in business contract related cases. The robot was claimed to outperformed them with accuracy level beyond 90 (Lu & Hu, 2019). These adventures of AI in law gives prospect of what AI can do even Nigeria.

However, the impact of technology and AI was not much felt during COVID-19 in Nigeria. The shutting down of courts stalled trials and reduced the rate of case disposal. The judicial system being heavily dependent on physical convergence of litigants, their lawyers and judge was adversely affected as a result of a halt in physical gathering (Olugasa & Davies, 2022). This setback was a lack of necessary ICT structures and software to drive a judicial system in a digital area. As the world gets into a new area of post covid-19, there is a need for innovative use of technology and Artificial Intelligence (AI) to drive judicial sector to achieve efficiency. It will speed up decision making process for the judges and increase case disposal rate

II. Impact of Covid-19 Pandemic on Judiciary

The COVID-19 pandemic had a major impact on Nigeria's judicial system, which changed how courts function. Yet, the epidemic also offered chances for the adoption of cutting-edge procedures and technologies that could eventually boost the effectiveness and accessibility of the legal system.

One significant effect is the delay in dispute resolution caused by the closing of courts to stop the spread of COVID-19, papers reported a surge in certain type of cases like contract (Ahmed, 2021; Arinze-Umobi & Okonkwo, 2021; Ojienda & Adude, 2021; Prescott et al., 2017) and domestic violence (Ahmed, 2021). This led to increase in backlog of cases and suffering for those with urgent cases.

Despite efforts to conduct court processes online, there were difficulties in cross-examining witnesses and determining the veracity of evidence virtually (Olugasa & Davies, 2022). The current level of technology and necessary software that can fast track this development are not adopted yet, affecting the effective use of technology in legal system. The level of connectivity in the country also played a significant role in access to justice, many Nigerians were unable to access to justice due to lack of technology or internet connectivity (Olugasa & Davies, 2022).

III. Smart Courts for Post Covid-19 Courts in Nigeria

Smart courts will require the use of intelligent software as a means of filing cases electronically and will also involve use of AI to support systems for decision-making both for judges and the lawyer. Large volume of files and documents are file in many cases, requiring a judge to read and study much before passing judgments (Reiling, 2020). This reason alone can be the cause to delay in many cases.

Limited use of online technology also greatly impacted the progress of trials during the outbreak of Covid-19 in Nigeria. Though technologies like video conference such as zoom, Skype were used in trying remote proceedings (Olugasa & Davies, 2022), they were grossly inadequate for various areas of court processes. They lack capacity for file documents and other essential cases management systems that will enhance proper documentation. Some areas that software technologies can judiciary are listed below.

3.1 Case Management systems

ICT can help courts manage record more effectively by automating procedures such as case scheduling and hearings, case file management, and case progress tracking (Prescott, 2017, Keong, 2017). CMS are also useful for creating reports. Several jurisdictions, such as the United States, use Odyssey by Tyler Technologies, JIMS by Journal Technologies, CMS by Thomson Reuters, and Court Explorer to speed up electronic filing, jury selection, and other administrative procedures. Similar technologies, such as JUSTIS and e-Curia, are utilized in courts in India, China, and Europe.

Sometimes, cases are never called or mention for years due to manual scheduling and human error. Therefore, case management is a crucial step in the administration of justice (Ojienda & Adude, 2021; Sangwan & Navet, 2021) and useful for assessing the speed and progress of cases and may even prevent cases from going uncalled for a long period of time.

3.2 Electronic filing Systems

Electronic filing systems, which allow parties to electronically file legal papers and court forms, are becoming increasingly prevalent in the judiciary (Emezue, 2020; Fenwick & Brownstone, 2002). It eliminates the need for paper filing and saves time. During Covid-19, the Chinese Judgments Online, China Legal Aid Online, and other related platforms were used to file new cases and make court pronouncements available online even before the pandemic. PACER (Public Access to Court Electronic Records) and CM/ECF (Case Management/Electronic Case Files) are also popular e-filing systems in the United States.

The Chief Justice of Nigeria and other state chief judges instructed Nigerian courts to file cases using WhatsApp, email, or Dropbox during the pandemic (Olugasa & Davies, 2022). This approach of case filling is out of step with modern technologies. Though a platform for standard case filling has been implemented in a few states in Nigeria, including Ogun, Rivers, and FCT (Olugasa & Davies, 2022), however, in state without portals to file cases, the correct files cannot be posted in the method of practice guidance for the judiciary.

Implementation of proper e-filling system is crucial in all state in Nigeria to forestall issues of missing cases or people being deny justice because they are unable file case within stipulated time by law. It also allows people to file cases without leaving the houses and reduce the risk of travelling.

3.3 Remote Hearing

In the wake of COVID-19, videoconferencing technology gained popularity among various organizations. It became a tool for judges in many countries like UK, USA, India and China to hold hearings and trials remotely (Keong, 2017). It eliminates the need for in-person appearances by connecting courts with parties, witnesses, and legal professionals.

Webex, and Microsoft Teams are three of the more popular video conferencing programs used in some countries including Nigeria (Olugasa & Davies, 2022), however, there is need for standard organizational-based software which Nigeria government will develop and continue to upgrade its functionality to suite its judiciary.

3.4 Digital Evidence management

Nigeria judiciary do not at the presents seem to have technology to handle digital evidence, however, digital evidence is becoming more important in the court system as a result of technology's continued advancement (Kessler, 2010). Effective management and storage are required as they gradually enter court cases. By offering tools for storing, indexing, and searching electronic documents and other types of digital evidence, ICT can assist courts in managing electronic evidence more efficiently. Some jurisdictions already employ programs like Trial Director and Sanction, Presentation Assistant, and Prezi for this reason.

3.5 Online Dispute Resolution:

Nigeria government have some dispute resolution departments in some agencies, however, the need for in-person appearance made them ineffective during the lockdown. Online dispute management and resolutions are very vital in the current era of technology and post Covid-19 Nigeria (Arinze-Umobi & Okonkwo, 2021; Zeleznikow, 2017). Though there is a concept of mobile court in Nigeria, it is completely a physical court which can change its location of sitting.

As the number cases are growing as a result of human population, some jurisdictions are testing online dispute resolution (ODR) systems, which leverage ICT to facilitate conflict settlement without necessity of in-person court appearances (Rule, 2020; Mania, 2015). ODR has developed as a viable alternative to regular judicial proceedings. Prior to COVID -19 era, China initiated the development of online courts in 2015 to trial specific sorts of cases, these efforts were especially important during COVID-19 when physical appearance could not possible.

3.6 Digital Transcription Tools:

For proper archiving and analysis, digital transcription tools are used to convert audio recordings of court proceedings. This software enables the legal team to concentrate on other crucial tasks and aids in distant proceedings. One crucial aspect of need in Nigeria is to develop digital transcription tools for the local languages, this tools will be essential to transcribe witnesses testimony in local languages.

IV. Artificial Intelligence (Ai) In Judicial System

The major input of AI in courts is to assist judges in decision making and to reduce a surge in the number of cases that many come to court (Sanctis, 2021; Wang, 2020). Several application of AI in court processes has shown remarkable success in legal practice (Re & Solow-niederman, 2019). AI has found its way into judiciary system (Giuffrida et al., 2018; Godwin, 2021; Journal & Sourdin, 2018; Kioko et al., 2022; Xu, 2022), and has transformed justice system in the form of an "assistant" judge which complements the efforts of judges to improve accuracy of judgements (Xu, 2022).

Some may believe that algorithms do not possess the capacity for fairness and proper interpretation of law or explanation of decision made (Soukupová, 2021), however, AI can be thought of as an implementation of codified justice system that could bring about fairness and reduction of arbitrariness in justice system (Re & Solow-niederman, 2019). It has capacity to transform legal system by predicting the outcomes of court decisions based on past judgements (Kauffman & Soares, 2020). With Artificial Intelligence, possibility of efficiency in judicial system with high level of predictability and accuracy of case outcome exist (Mccarthy, 2020). It will also improve the quality of decisions and reduce time taken during trials and proceedings. There are simple and complex cases which into which AI can support judge in different front Like examination of evidence during proceeding and the generation of elemental adjudication instruments in a short period (Kauffman & Soares, 2020; Xu, 2022).

4.1 Specific areas of AI application in judiciary

4.1.1 Predicting outcome of a case

Based on past data, AI can assist lawyers in predicting the result of legal disputes. This can enable them to decide more intelligently whether to settle a lawsuit or proceed to trial. Prediction capacity of AI application has been prevalent in many areas of science before now and has been of great power for helping in strategic making decisions, however, it's starting to be implemented in areas that do not seem to follow precise rule.

AI software developed in Texas University using US Supreme Court judgment from 1791 to 2015 was able to predict Supreme Court judges' votes and decision with over 70% accuracy. This level of accuracy is said to be higher than the average human 66% predictive accuracy of jurists (Xu, 2022). Due to frequency of pattern of certain cases, a relatively large proportion of routine cases have a predictable outcome. In those cases, the court rulings are relevant for input data into AI-based application and future predictions judgment of other cases (Mccarthy, 2020; Reiling, 2020). Users of such AI software can easily predict the chances of winning a particular case even before filling it and make right decision.

4.1.2 Evidence Analysis

One of the biggest challenges of digital era is the possibility of digital manipulation and faking of evidence. A number of deep learning and AI applications exist to handle the complexity and evidence management (Wang, 2020). The Shanghai judiciary developed the Intelligent Auxiliary Case Handling System for Criminal Cases. The software manages case through Optical Character Recognition (OCR) and Natural Language Processing (NLP). The software is developed to map evidence to appropriate crimes and to review evidence to find probable contradictions. Variations and relevance of the evidence to the case can even be establish using the software. In the era of remote court proceeding, AI stands as an important component which will help judges to determine genuineness of evidence.

4.1.3 Extracting Patterns in Case Files

Pattern recognition is one powerful area that AI has thrived. Machine learning algorithms are employed to recognize patterns in a large volume of data (Karatzinis & Boutalis, 2021). Sorting several cases into categories or finding pattern in a case with large volume of information could be a daunting task; however, AI application like 'eDiscovery' being used in American courts performs intelligent analysis of case files before the start of a court procedure (Reiling, 2020). eDiscovery uses machine learning to extract the relevant parts from a large amount of information (Kauffman & Soares, 2020). This help to focus on the case more precisely.

'SOCRATES' is a similar AI based application which was trained from over 300,000 previous court judgments (Sanctis, 2021). The AI software scan and read through all the case file and cluster them so that they can be judged in blocks. It functions to screen out cases that are outside the jurisdiction of the court so that the court time won't be wasted. Cases similar to previously judged cases are return to the lower courts so that the previous judgments of the court on such cases can be apply on them.

4.1.4 Sentence preparation

Preparing court sentence can be challenging and time consuming due to large document to read, however, AI can speed up this process. A good example is VICTOR, an AI software used by Brazilian Supreme Court Which read all appeal cases filed at the Supreme Court and identify which ones are linked to certain repercussion. The software was claimed to have speed up the work of the judges. It is also being proposed as a support system to judges in the lower courts to reduce number of cases that gets to appeal court (Sanctis, 2021). With this very Machine learning application, the number of possible cases will be reduced with increase in processing of appeal. The software has resolved about 10,000 extraordinary appeals that should have reach the supreme (Sanctis, 2021)

4.1.5 Sentiment Analysis and Natural Language Processing

AI can also be used to analyze legal texts to identify relevant legal principles, arguments or sentiment of a particular individual. This can help to establish motive or intent in a case. The judges will also have more informed decisions from such analysis. A close case of this research in UK by European Court of Human Rights detects human right infringement base on choices and language usage (Mccarthy, 2020). The success rate accomplished by the software is close to 80%. With this remarkable success, it can be a good support system for judges handling human right cases.

4.1.6 Legal counsel

Legal advice helps litigants to take decision of on resolution of cases and reduce the need to court cases. A number of AI software are being developed to help the public in such instances so that if proper decision and resolution occur, there won't be need to file for court case. Examples are AI applications developed to help the District Court of East Brabant in the Netherlands in administrative handling of traffic violation cases. A similar software developed for advising parties on cases relating to housing dispute and accidents from collisions is being used by Civil Resolution Tribunal (CRT) Columbia, Canada (Reiling, 2020). The software offers legal advice and information to the public on paths to resolve dispute. This software are very vital in resolving trivial cases.

4.1.7 Legal research

Legal research tools allow legal professionals to search and access legal resources. By combining machine learning and natural language processing, AI-powered software can spot patterns and insights in vast amounts of legal data that would be challenging for humans to discover on their own.

V. Ethical Concern Over Use of Ai in Court

Though the use of AI in legal system has prospect in empowering judiciary for speedy dispensation of justice; however, there are concern on the use of AI in court on challenges of infringement of peoples right due to result of data analysis (Moroianu et al., 2019). A number of issues were outlined by (Reiling, 2020) which include fairness of AI software to all parties during trials, respect to fundamental human right of all parties and non-discrimination. The problem of taking responsivity for whatever decision an AI system may take is also a very crucial aspect of law, an AI system cannot be sued for damages if it offers wrong advice or the service result damages.

VI. Conclusion

As populations grow, crisis and litigations also grow among humans. This fact has led to an ever-increasing surge of cases that come before judicial officers for adjudication. However, Judiciary in Nigeria and globally are challenged daily with large number of cases beyond the available manpower, this has led to many not having their cases had for years. However, there is need for collaborative research among scientist from various in other to produce efficient systems for legal service. The current level of over dependence on manual system and physical proceedings are also a major challenge. Implementation of intelligent software and use of appropriate technology can improve access to justice, speed up the work and reduce the burden on the judicial staff. It will improve the quality of judgment and reduce arbitrariness of human judges.

Reference

1. Ahmed, S. (2021). New legal challenges post covid with specific reference to Indian contract Act. *Palarch's Journal Of Archaeology Of Egypt/Egyptology*, 18(9), 96–102.
2. Arinze-Umobi, C. N., & Okonkwo, I. T. (2021). Alternative dispute resolution practice in nigeria and the effect of covid-19 pandemic. *IJOLACLE*, 2(May), 82–85.
3. Contini, F. (2020). Artificial Intelligence and the Transformation of Humans, *Law and Technology Interactions in Judicial Proceedings*. *Law, Technology and Humans*, 2(1), 4–18. <https://doi.org/10.5204/lthj.v2i1.1478>
4. Emezue, C. (2020). Digital or digitally delivered responses to domestic and intimate partner violence during COVID-19. *JMIR public health and surveillance*, 6(3), e19831.
5. Fenwick, W. A., & Brownstone, R. D. (2002). Electronic Filing: What Is It-What Are Its Implications. *Santa Clara Computer & High Tech. LJ*, 19, 181.
6. Giuffrida, I., Lederer, F., & Vermerys, N. (2018). A Legal Perspective on the Trials and Tribulations of AI : How Artificial Intelligence , the Internet of Things , Smart Contracts , and Other Technologies Will Affect the Law. *Case Western Reserve Law Review Volume*, 68(3), 747–781.
7. Godwin, J. (2021). Position of Artificial Intelligence in Justice System : Justice of the Future. 1–15.
8. Journal, U. L., & Sourdin, T. (2018). Judge v Robot ? Artificial intelligence and judicial decision-making. *UNSW Law Journal*, 41(4), 1114–1133.
9. Karatzinis, G. D., & Boutalis, Y. S. (2021). Fuzzy cognitive networks with functional weights for time series and pattern recognition applications. *Applied Soft Computing*, 106, 107415. <https://doi.org/10.1016/j.asoc.2021.107415>
10. Kauffman, M. E., & Soares, M. N. (2020). AI in legal services : new trends in AI-enabled legal services. *Service Oriented Computing and Applications*, 14(4), 223–226. <https://doi.org/10.1007/s11761-020-00305-x>
11. Keong, G. C. (2017). Judicial Reforms through the Use of Technology in Malaysia. *European Academic Research*, 5(1), 399–409.
12. Kessler, G. C. (2010). Judges' awareness, understanding, and application of digital evidence. *Nova Southeastern University*. Retrieved from https://nsuworks.nova.edu/cgi/viewcontent.cgi?article=1195&context=gscis_etd/ on 22/03/2023
13. Kioko, P. M., Booker, N., Chege, N., & Kimweli, P. (2022). The Adoption of Artificial Intelligence in Newsrooms in Kenya: a Multi-case Study. *European Scientific Journal, ESJ*, 18(22), 278. <https://doi.org/10.19044/esj.2022.v18n22p278>
14. Lu, H., & Hu, T. (2019). Study on the Influence of Artificial Intelligence on Legal Profession. *Dvances in Economics, Business and Management Research*, 110(Emle), 964–968.
15. Mccarthy, J. (2020). Role of Artificial Intelligence in Transforming the Justice Delivery System in COVID 19 Pandemic. *International Journal on Emerging Technologies*, 11(3), 344–350.
16. Moroianu, I., Tescu, Z. L. Ă., Emanuel, P., & Tescu, Z. L. Ă. (2019). of Law Implementation of the european ethical

charter on the use of artificial intelligence in judicial systems and their environment. Supplement of Law Review, 237–242.

17. Ojienda, T., & Adude, L. M. (2021). Access to Justice in the era of COVID-19: Adaptations and Coping Mechanisms of the legal Services Industry in Kenya. *Journal of Conflict Management and Sustainable Development*, 6(2), 1–44.
18. Olugasa, O., & Davies, A. (2022). Remote Court Proceedings in Nigeria: Justice Online or Justice on the Line. *International Journal for Court Administration*, 13(2), 1–17. <https://doi.org/10.36745/ijca.448>
19. Prescott, J. J., Bulinski, M., Rozema, K., Bouzat, F., Bridges, G., Epstein, T., Gallmeyer, J., Hogin, J., Kim, S., Moore, M., Perry, M., Piligian, J., & Rouillard, A. (2017). Improving Access to Justice in State Courts with Platform Technology, 70 *Vanderbilt Law Review*. *Vanderbilt Law Review*, 70(6), 1995–2050.
20. Re, R. M., & Solow-niederman, A. (2019). Developing Artificially Intelligent Justice. *Stan. Tech. L. Rev.*, 242(2), 242–289.
21. Reiling, A. D. D. (2020). Courts and Artificial Intelligence. *International Journal For Court Administration*, 11(2). <https://doi.org/https://doi.org/10.36745/ijca.343>
22. Sanctis, F. M. De. (2021). Artificial Intelligence and Innovation in Brazilian Justice. *International Annals of Criminology*, 1–10. <https://doi.org/10.1017/cri.2021.4>
23. Sangwan, S., & Navet, R. S. (2021). digitalization of legal services (No. 1).
24. Soukupová, J. (2021). Ai-based legal technology: a critical assessment of the current use of artificial intelligence in legal practice. *Masaryk University Journal of Law and Technology*, 15(2), 279–300. <https://doi.org/10.5817/MUJLT2021-2-6>
25. Wang, R. (2020). Legal technology in contemporary USA and China. *Computer Law & Security Review*, 39, 1–20. <https://doi.org/10.1016/j.clsr.2020.105459>
26. Xu, Z. (2022). Human Judges in the Era of Artificial Intelligence : Challenges and Opportunities. *Applied Artificial Intelligence*, 36(1). <https://doi.org/10.1080/08839514.2021.2013652>
27. Zeleznikow, J. (2017). IJCA - Can Artificial Intelligence And Online Dispute Resolution Enhance Efficiency And Effectiveness In Courts. 8. <https://doi.org/10.18352/ijca.223>